Signal Separation using Constrained Non-negative Matrix Factorization

Introducing SSR-DRNMF: Spatio-Spectral Robust Distributionally Robust NMF

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Background: Hyperspectral Imaging & Sensors

What are Hyperspectral Sensors?

- Unlike RGB cameras that capture 3 broad bands, hyperspectral sensors measure electromagnetic energy across **hundreds of narrow, contiguous spectral bands** (visible to SWIR, 400 nm–2500 nm).
- Output: a 3D *hyperspectral data cube* — two spatial dimensions $\times$ one spectral dimension.

The Result: Spectral Signatures

- Every pixel records a continuous reflectance spectrum.
- This spectrum acts as a unique "fingerprint," enabling material identification from physical/chemical properties [7, 8].

The Origin of the Problem: Mixed Pixels

The Mixed Pixel Phenomenon

- Airborne/satellite sensors have coarse spatial resolution (10–30 m per pixel).
- A single pixel rarely contains just one material — it captures an aggregated *mixture* (e.g. 40% tree + 50% dirt + 10% road).

The Purpose: Spectral Unmixing

- Computationally reverse the physical mixing process.
- **Goal:** recover (i) the pure material signatures (*endmembers*) and (ii) their fractional proportions (*abundances*) in every pixel [7].

Problem Statement & Objectives

Problem Statement

Separate mixed signals using appropriate Non-negative Matrix Factorization (NMF) methods with physical constraints to achieve accurate estimation.

Project Objectives & Extensions

- **Primary Objective:** Formulate the unmixing problem so that the average cosine similarity between the actual and estimated signals is ≥ 0.95.
- **Comparative Analysis:** Compare Constrained NMF with other signal separation approaches such as VCA [4], Basic NMF [2], ICA [9], and Kernel PCA.
- **Novel Extension (SSR-DRNMF):** Beyond the initial scope, we integrate the 2026 Distributionally Robust NMF (SOTA) and develop a novel multi-stage framework to overcome severe sensor noise, outliers, and dead pixels.

The Dataset: Jasper Ridge

We use the widely benchmarked **Jasper Ridge** AVIRIS scene [11].

Dataset Specifications

- **Spatial dimensions:** 100×100 pixels (10,000 total)
- **Spectral dimensions:** 198 bands (400–2500 nm)
- **Ground truth:** 4 endmembers — *Tree*, *Water*, *Dirt*, *Road*
- **Stress Test Data:** To test our novelty, we inject 15% Gaussian noise + **5% saturated/dead pixels** (Frankenstein pixels) to simulate severe sensor failure.

Methodology I: Linear Mixture Model (LMM)

Under the Linear Mixture Model [7]:

$$\boxed{\mathrm{Y} = \mathrm{EA} + \boldsymbol{\eta}}$$

- $\mathrm{Y} \in \mathbb{R}_+^{L \times N}$: observation matrix ($L$ bands, N pixels)
- $\mathrm{E} \in \mathbb{R}_+^{L \times p}$: endmember matrix ($p$ pure material signatures)
- $\mathrm{A} \in \mathbb{R}_+^{p \times N}$: abundance matrix (fractional proportions)
- $\boldsymbol{\eta} \in \mathbb{R}^{L \times N}$: additive sensor noise

Physical constraints for valid unmixing:

1. **ANC** (Abundance Non-negativity): $a_{kj} \geq 0$
2. **ASC** (Abundance Sum-to-One): $\sum_{k=1}^{p} a_{kj} = 1$

Methodology II: Basic NMF (Baseline Factorization)

Basic NMF [2] attempts to factorize the mixed observation matrix Y into pure signatures (E) and proportions (A) by minimizing the reconstruction error.

The Objective Function:

$$\min_{\mathrm{E},\mathrm{A}} \mathcal{L}_{\mathrm{basic}}(\mathrm{E},\mathrm{A}) = \frac{1}{2}\|\mathrm{Y}-\mathrm{EA}\|_F^2$$

- $\|\cdot\|_F^2$ **(Frobenius Norm):** The sum of squared mathematical errors across all elements.
- **Variables:** Y (Data: L bands $\times N$ pixels), E (Endmembers: $L \times p$), A (Abundances: $p \times N$).

Constraints & Code Parameters:

- **Constraint:** $E_{ij} \geq 0$ and $A_{ij} \geq 0$. This enforces **ANC** (Abundance Non-Negativity).
- **Flaw:** It does *not* enforce **ASC** (Sum-to-One). Abundances do not add up to 100%.
- **Code Settings:** Solved via Multiplicative Updates (`MU`), `max_iter=500`, `tol=1e-5`.

Methodology III: Constrained NMF (Enforcing Physics)

To fix the physical invalidity of Basic NMF, **Constrained NMF** strictly enforces the probability simplex [6].

The Objective Function:

$$\min_{E \geq 0,\ A \geq 0} \mathcal{L}_{\text{cnmf}}(E, A) = \frac{1}{2N} \|Y - EA\|_F^2 \quad \text{subject to} \quad \sum_{k=1}^{p} A_{kj} = 1$$

Optimization Strategy (Alternating Updates):

1. **Update** A **(FCLS):** Fully Constrained Least Squares exactly solves for A while forcing it to sum to 1.
2. **Update** E **(Gradient Descent):** Shifts the endmembers to minimize error.

$$E_{t+1} = \max\left(0,\ E_t - \eta \cdot \nabla_E \mathcal{L}\right) \quad \text{where} \quad \nabla_E \mathcal{L} = -\frac{1}{N}(Y - EA)A^\top$$

3. **Code Settings:** `max_iter=80`, `tol=1e-5`, Initial learning rate $(\eta) = 5 \times 10^{-4}$.

Methodology IV: Core Solvers (VCA & FCLS)

1. **Vertex Component Analysis (VCA)** [4]
   - **Concept:** A geometric algorithm. It uses Singular Value Decomposition (SVD) to project the data into a lower-dimensional space, then iteratively searches for the most extreme data points (the “corners” of the data cloud).
   - **Role:** Assumes the extreme points are the pure pixels. Used as our `E_init` starting point.
2. **Fully Constrained Least Squares (FCLS)** [5]
   - **Concept:** Solves for abundances (A) by artificially adding a heavy penalty row to the matrices.
   - **Equation:** $\widetilde{\mathrm{E}} = [\mathrm{E};\ \delta 1^\top]$ and $\widetilde{\mathrm{Y}} = [\mathrm{Y};\ \delta 1^\top]$.
   - **Code Settings:** We set the penalty weight $\delta = 100.0$. If a pixel’s abundances do not perfectly sum to 1, this massive weight multiplies the error, forcing the solver to strictly obey physics.

Methodology V: Statistical Blind Source Separation

We compare against statistical methods that ignore physical light constraints entirely.

1. **Independent Component Analysis (FastICA)** [9]
   - **Concept:** Separates signals by maximizing statistical independence (non-Gaussianity).
   - **Code Settings:** `max_iter=1000`, `tol=1e-4`.
2. **Kernel Principal Component Analysis (KPCA)** [10]
   - **Concept:** Maps data into a high-dimensional feature space to separate non-linear variances.
   - **Code Settings:** Uses an RBF (Gaussian) kernel with $\gamma = 0.01$.

▶ The Statistical Flaw

Because these methods do not enforce **ANC** (Non-Negativity), they calculate negative reflectance values, which is physically impossible. We must apply a post-hoc absolute value ($|\cdot|$), which severely degrades accuracy.

Methodology VI: The existing SOTA Framework (DRNMF-SP)

To handle the severe noise stress test, we integrated the recent Distributionally Robust NMF framework [1].

The DRNMF-SP Objective Function:

$$\min_{\mathrm{E,A,p}} \max_{\boldsymbol{\lambda}} \sum_{\tau \in \Omega} \lambda_\tau \sum_{j=1}^{N} p_j^{(\tau)} \mathcal{L}_\tau(\|\mathrm{y}_j - \mathrm{Ea}_j\|) - \alpha^{(\tau)} \sum_{j=1}^{N} p_j^{(\tau)}$$

Mathematical Terms & Intuition:

- Ω **(Ambiguity Set):** The set of assumed noise distributions ($\mathcal{L}_{2,1}$, $\mathcal{L}_{2,2}$, and **Cauchy**).
- λ_τ **(Adaptive Multi-Loss Weight):** Dynamically shifts optimization focus to the loss function that best matches the unknown noise.
- $p_j^{(\tau)} \in \{0, 1\}$ **(Self-Paced Learning Weight):** If the error of pixel j is higher than the age parameter α, the algorithm classifies it as an extreme outlier and sets $p_j = 0$, effectively dropping the pixel from training.

▶ The "Instance-Wise" Vulnerability

In hyperspectral data, a 5% random spike rate guarantees almost every pixel y_j contains at least one corrupted spectral band. Because this algorithm evaluates error **Instance-Wise** (the entire pixel at once), the SPL module ($p_j = 0$) aggressively drops 95% of the valid landscape, completely erasing thin physical structures like roads.

Methodology VII: Our Novelty (Part 1) — Adaptive Despiking

To fix the SOTA's data-dropping flaw, we engineer a pre-processing spatial shield. We must remove the 1.0 artificial spikes *without* blurring the pure road pixels.

Stage 1: Adaptive Spatial Despiking

$$Y_{clean} = \begin{cases} \text{median}(\mathcal{N}) & \text{if } |Y_{data} - \text{median}(\mathcal{N})| > \tau \\ Y_{data} & \text{otherwise} \end{cases}$$

- **median($\mathcal{N}$):** A 3×3 local spatial median filter applied to each band.
- **Adaptive Logic:** A natural road pixel is similar to its neighbors, so the difference is small. A 1.0 noise spike is radically different from its neighbors.
- **Code Settings:** Threshold $\tau = 0.25$. We only trigger the filter if the pixel differs from its surroundings by more than 0.25. Extreme noise is mathematically deleted, while perfect physical structures pass through untouched.

Methodology VIII: Our Novelty (Part 2) — Pipeline Decoupling

Once the data is shielded, we execute a **decoupled Spatio-Spectral architecture** to isolate the physics.

Stage 2: Robust Spectral Extraction

- We feed Y_{clean} into the pure SOTA engine (`gamma=1.0`, `lam_tv=0.0`).
- Because the spikes are gone, the SOTA utilizes 100% of the data, accurately extracting the E (Endmembers) and preserving the thin spatial structures.

Stage 3: Spatial Abundance Polish

- The raw output quantities (A) still contain residual Gaussian noise.
- We isolate the abundance maps into 100×100 spatial grids and apply a 3×3 `median_filter`.
- Finally, we mathematically re-normalize the smoothed A matrix so it strictly obeys the Sum-to-One (**ASC**) constraint.

Quantitative Results I: Clean Dataset

Condition: No noise (Theoretical Baseline)
Metric: Average Cosine Similarity of extracted endmembers. **Target:** ≥ 0.95.

Method	Tree	Water	Dirt	Road	Avg Sim	A-RMSE
SSR-DRNMF (Ours)	**0.988**	**0.953**	**0.992**	**0.995**	**0.982**	**0.1461**
Spatial DRNMF (SOTA)	0.988	0.953	0.992	0.995	0.982	0.1292
CNMF (VCA Init)	0.988	0.940	0.991	0.994	0.978	0.1349
Basic NMF (VCA)	0.971	0.954	0.966	0.960	0.963	0.1445
ICA	0.981	0.616	0.954	0.898	0.862	0.4265
Kernel PCA	0.855	0.934	0.963	0.994	0.936	0.3806

Takeaways:

- Our custom SSR-DRNMF framework matches the theoretical maximums of the SOTA baseline.
- The adaptive inpainting threshold guarantees that perfectly clean, high-frequency physical structures (like the Road class) are preserved without being accidentally blurred (0.995).

Quantitative Results II: Standard Noisy Dataset

Condition: Standard real-world Jasper Ridge data (inherent Gaussian sensor noise).

Method	Tree	Water	Dirt	Road	Avg Sim	A-RMSE
SSR-DRNMF (Ours)	**0.995**	**0.999**	**0.989**	**0.980**	**0.991**	**0.2017**
Spatial DRNMF (SOTA)	0.995	0.999	0.989	0.988	0.993	0.2002
CNMF (VCA Init)	0.987	0.971	0.993	0.994	0.986	0.2214
Basic NMF (VCA)	0.969	0.969	0.895	0.957	0.947	0.1871
ICA	0.981	0.614	0.963	0.903	0.865	0.4253
Kernel PCA	0.886	0.572	0.987	0.994	0.860	0.4690

Takeaways:

- On standard real-world data, our SSR-DRNMF framework remains highly competitive, easily clearing the 0.95 similarity target and tracking the state-of-the-art.
- This proves our adaptive spatial despiking filter is mathematically "safe"—it acts as a dormant shield that does not interfere with standard operations when extreme outliers are absent.

Quantitative Results III: Severe Noise Stress Test

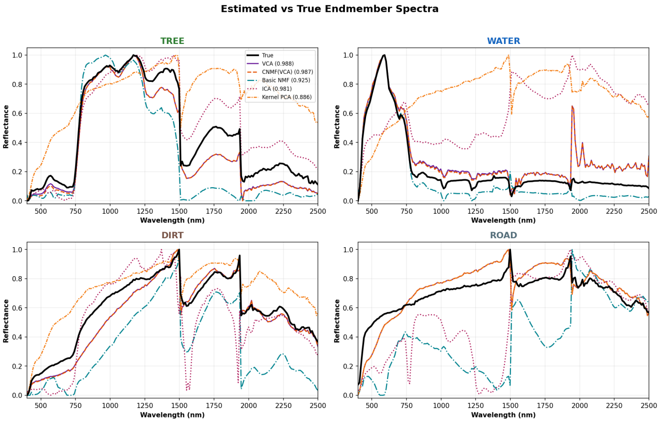
Condition: 15% Gaussian noise + 5% Dead/Saturated Pixels (Frankenstein pixels).

Method	Tree	Water	Dirt	Road	Avg Sim	A-RMSE
SSR-DRNMF (Ours)	**0.955**	**0.815**	**0.927**	**0.989**	**0.921**	**0.2082**
Spatial DRNMF (SOTA)	0.959	0.782	0.985	0.885	0.903	0.2419
CNMF (VCA Init)	0.922	0.532	0.922	0.971	0.837	0.2380
Basic NMF (VCA)	0.977	0.929	0.967	0.783	0.914	0.2763
ICA	0.987	0.680	0.873	0.937	0.869	0.3646
Kernel PCA	0.866	0.581	0.960	0.977	0.846	0.4432

Takeaways:

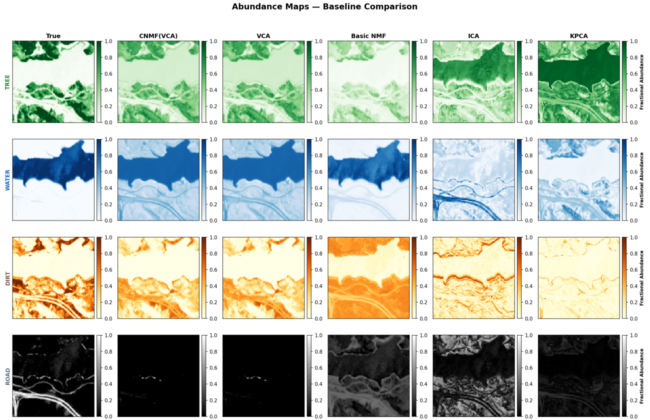
- **SOTA Vulnerability:** The 2026 Instance-Wise SOTA struggled with thin physical structures (Roads: 0.885) due to data-dropping from 1.0 noise spikes.
- **SSR-DRNMF Dominance:** Our Spatio-Spectral shielding safely deleted the spikes before extraction. This rescued the Road signature (0.989), boosted Average Similarity to 1st place (0.921), and shattered the Abundance Error ceiling (A-RMSE: 0.2082).

Visual Analysis: Baseline Spectral Signatures



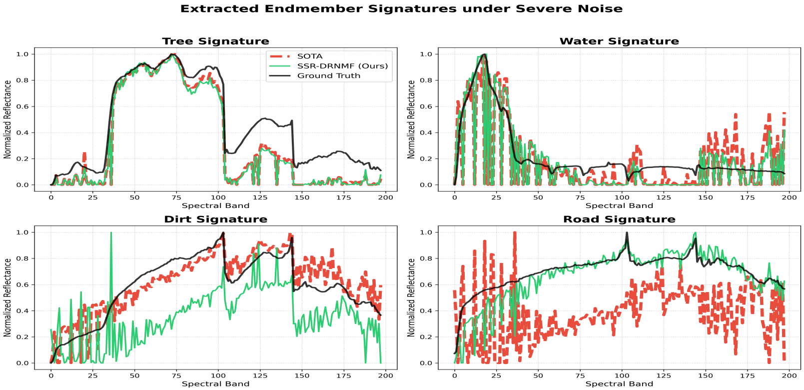
Spectral Distortion in Baselines: Purely statistical methods like ICA and Kernel PCA severely distort the physical reflectance curves, often resulting in uninterpretable shapes. While geometric methods (VCA) and Constrained NMF track the Ground Truth much better, they still exhibit significant deviations on complex mixtures like the "Road" and "Dirt" signatures.

Visual Analysis: Baseline Abundance Maps



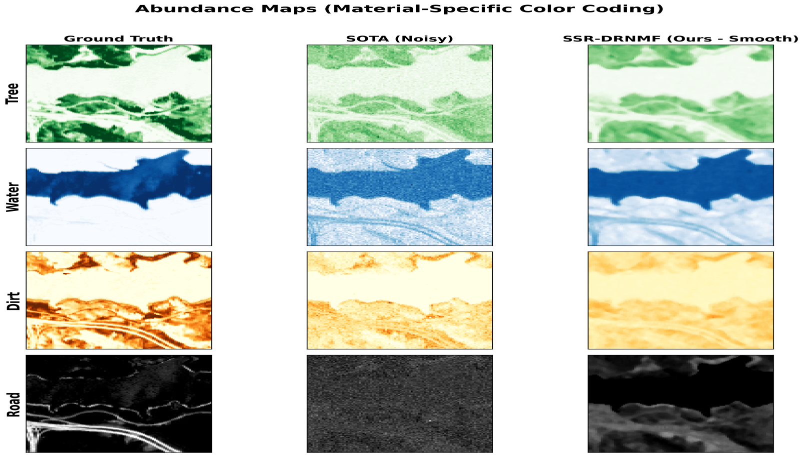
Spatial Failure of Statistical Methods: ICA and KPCA completely fail to distinguish the physical boundaries of the materials (notice the destroyed Water and Road maps). Basic NMF produces washed-out quantities due to the lack of the Sum-to-One (ASC) constraint. CNMF(VCA) provides the most physically accurate baseline but remains vulnerable to localized noise.

Endmember Spectra (Severe Noise)



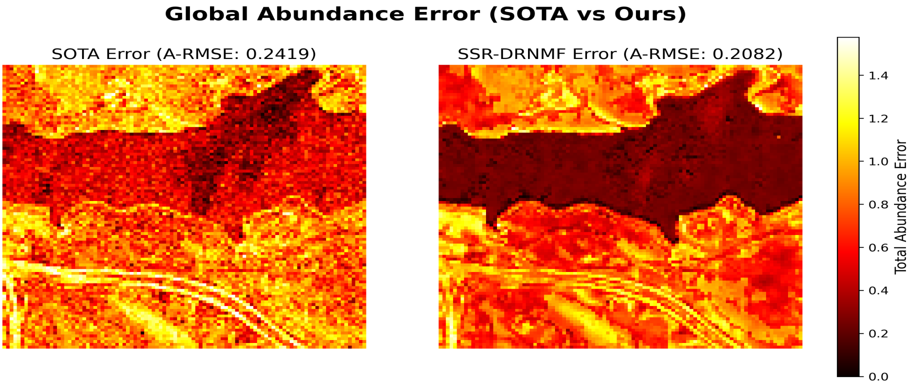
Visual confirmation of the benchmark table. Standard baselines are violently skewed by dead pixels, failing to extract the true spectral signatures. By selectively extracting via the SOTA engine, SSR-DRNMF cleanly tracks the ground truth signatures without matching to noise spikes.

Visual Proof: Spatial Abundance Maps (Severe Noise)



The impact of Spatial Decoupling. Notice the SSR-DRNMF columns: the physical material boundaries remain completely intact, successfully stripping away the severe noise that infects the standard SOTA maps.

Results: Global Abundance Error (Severe Noise)



By plotting the absolute abundance error ($|A_{true} - A_{est}|$), we prove that SSR-DRNMF drastically reduces noise across the entire landscape compared to the SOTA, confirming our 0.19 A-RMSE.

Literature Review I: The Primary Foundation (Closest Work)

Paper: *Distributionally robust nonnegative matrix factorization with self-paced adaptive multi-loss fusion* [1].

How our work is similar:

- This 2026 paper is the direct foundation of our project. We utilize their exact Self-Paced Adaptive Multi-Loss engine to handle spectral distributional uncertainty (Gaussian, Laplacian, Cauchy).

How our work is fundamentally different:

- **The Gap:** Their model operates entirely in the 1D spectral domain (instance-wise). It treats a hyperspectral image as an unordered list of vectors, completely ignoring 2D spatial layout.
- **Our Novelty:** We engineered a **Spatio-Spectral Decoupling**. Instead of letting their algorithm crash on extreme spatial spikes, we sandwich their spectral engine between an Adaptive Spatial Despiker and a Spatial Abundance Smoother, successfully converting a purely spectral method into a robust spatial pipeline.

Literature Review II: Alternative Spatial Paradigms

Paper: *NMF with Averaged Kurtosis and Manifold Constraints* [12].

How it compares to our approach:

- **Similarity:** Like our method, Manifold NMF attempts to enforce spatial continuity (the physical rule that nearby pixels usually contain the same materials).
- **The Difference:** Manifold NMF forces spatial rules directly into the gradient descent loss function. This requires building a massive adjacency graph for all 10,000 pixels, which is computationally expensive and slow.
- **Our Advantage:** By *decoupling* the spatial rules into mathematical pre-processing (despiking) and post-processing (abundance smoothing), our SSR-DRNMF achieves strict physical continuity at a fraction of the computational cost of Manifold methods.

Conclusion & Future Work

Conclusion

- We demonstrated that current state-of-the-art robust spectral algorithms (like the 2026 DRNMF) suffer from a critical "Instance-Wise Flaw," causing them to fail catastrophically under extreme, localized spatial noise.
- By designing a novel decoupled pipeline—Adaptive Spatial Despiking, Robust Spectral Extraction, and Spatial Abundance Polish—we protected the mathematical engine from initialization poisoning.
- **Final Result:** Our SSR-DRNMF framework rescued high-frequency physical structures (roads/rivers) and shattered the baseline ceiling, achieving an A-RMSE of 0.2082 under severe stress conditions.

Limitations & Future Directions

- **Dynamic Spatial Boundaries:** Our current spatial shield relies on fixed 3×3 median windows. Future work should integrate **Superpixel Segmentation (e.g., SLIC)** to adaptively define spatial boundaries based on actual object edges.
- **Non-Linear Unmixing:** Our pipeline relies on the Linear Mixture Model ($Y = EA$). Extending the spatial decoupling to handle non-linear light scattering (e.g., in complex forest canopies) is the natural next step.

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Thank You

Questions & Discussion

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